Customer Shopping Behavior Analysis

1.PROJECT OVERVIEW

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2.TOOLS & TECHNOLOGIES USED

• Python
• Pandas
• NumPy
• PostgreSQL
• SQL
• Power BI
• Microsoft Excel
• Microsoft PowerPoint

3. DATASET SUMMARY

Rows: 3,900
Columns: 18

Key Features:

• Customer Demographics:
- Age
- Gender
- Location
- Subscription Status

• Purchase Details:
- Item Purchased
- Category
- Purchase Amount
- Season
- Size
- Color

• Shopping Behavior:
- Discount Applied
- Promo Code Used
- Previous Purchases
- Frequency of Purchases
- Review Rating
- Shipping Type

Missing Data:

• 37 missing values identified in the Review Rating column.

4. EXPLORATORY DATA ANALYSIS USING PYTHON

We began with data preparation and cleaning in Python:

- **Data Loading:**
  Imported the dataset using Pandas.
- **Initial Exploration:**
  Used df.info(), df.describe(), and value counts to understand the structure and characteristics of the data.
- **Missing Value Treatment:**
  Identified missing values and imputed Review Rating using the median rating of each product category.
- **Column Standardization:**
  Renamed column names into snake_case format for consistency and easier SQL querying.
- **Feature Engineering:**
  Created age_group for customer segmentation.
  Created additional derived variables to support customer behavior analysis.
- **Data Validation:**
  Checked for duplicates, inconsistencies, and redundant variables.
- **Database Integration:**
  Connected Python with PostgreSQL using SQLAlchemy and loaded the cleaned dataset into the database for SQL analysis.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	N
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	N
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	N
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	N
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	N
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	N
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	N

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

5. DATA ANALYSIS USING SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	[illegible]

Total rows: 839 Query complete 00:00:

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

revenue across subscription status.

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

of discounted purchases.

7. **Customer Segmentation** – Classified customers into New, Returning, and

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

Loyal segments based on purchase history.

8. **Top 3 Products per Category** – Listed the most purchased products within

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

each category.

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

6. KEY INSIGHTS

1. Clothing emerged as the highest revenue-generating category and also recorded the highest sales volume.
2. The majority of customers were non-subscribers, indicating an opportunity to improve subscription adoption.
3. Repeat customers demonstrated stronger purchasing behavior and higher engagement levels.
4. Customer spending patterns varied significantly across different age groups.
5. Discount campaigns influenced purchasing decisions for several products.
6. Certain products consistently received higher review ratings and purchase volumes, indicating strong customer preference.
7. Product categories exhibited different purchasing trends across customer segments.
8. Loyal customers contributed significantly to overall business performance.

7.DASHBOARD DEVELOPMENT IN POWER BI

An interactive Power BI dashboard was developed to visualize customer behavior and business performance.

Dashboard Components:

• **KPI Cards**

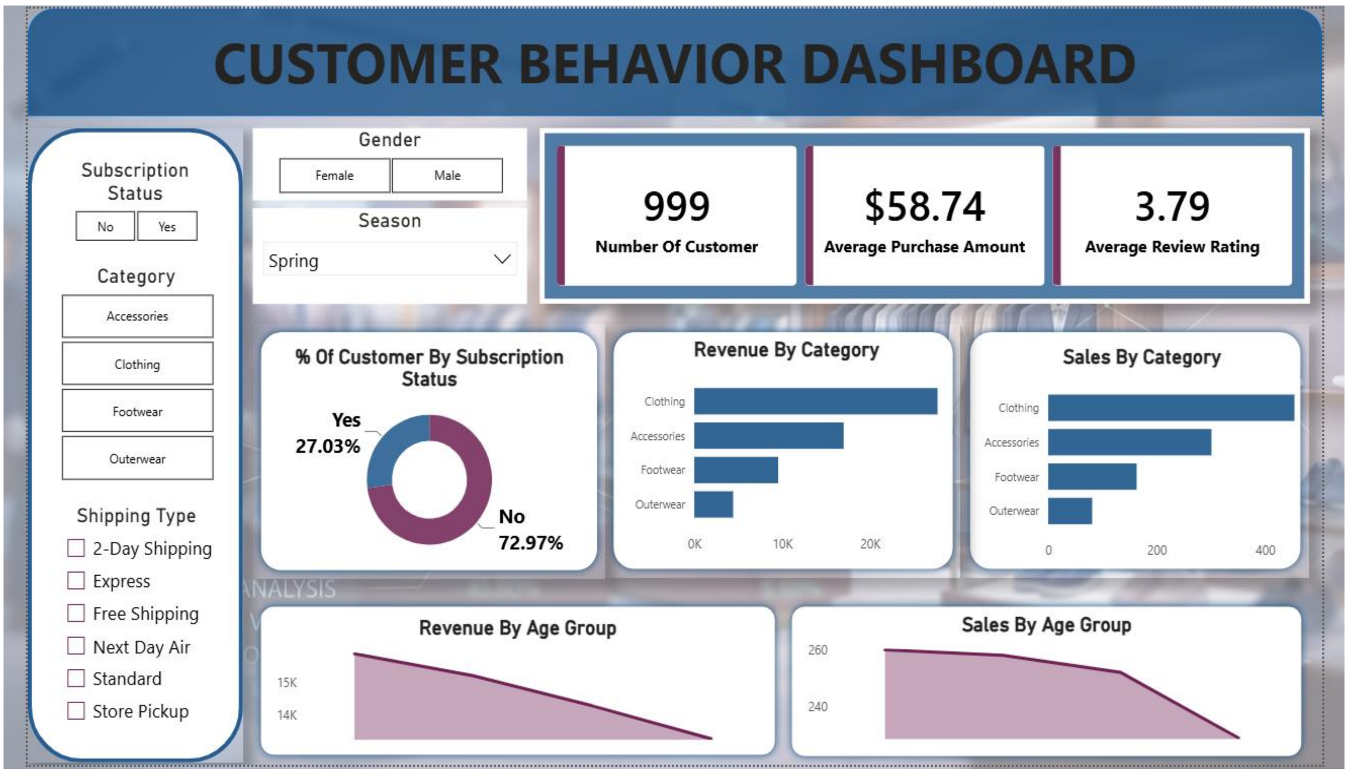
- Total Customers
- Average Purchase Amount
- Average Review Rating

• **Interactive Filters**

- Gender
- Category
- Subscription Status
- Season
- Shipping Type

• **Visualizations**
 - Revenue by Category
 - Sales by Category
 - Revenue by Age Group
 - Sales by Age Group
 - Customer Subscription Distribution

The dashboard enables users to dynamically explore customer trends and business performance across multiple dimensions.



8. BUSINESS RECOMMENDATIONS

• **Increase Subscription Adoption:**
 Offer exclusive discounts, loyalty rewards, and faster shipping options for subscribers.

• **Strengthen Customer Loyalty Programs:**
 Reward repeat buyers through tier-based loyalty programs and personalized promotions.

• **Optimize Discount Strategies:**
 Focus discount campaigns on products and customer segments that generate higher conversions.

• **Promote High-Performing Products:**

Highlight top-selling and highly rated products in marketing campaigns.

- **Target High-Value Customer Segments:**
  Develop targeted campaigns for age groups contributing the highest revenue.
- **Improve Customer Retention:**
  Encourage first-time customers to become repeat purchasers through personalized recommendations and offers.

9. CONCLUSION

This project demonstrates an end-to-end Data Analytics workflow involving data cleaning, preprocessing, SQL analysis, dashboard development, and business reporting.

Using Python, PostgreSQL, SQL, and Power BI, customer purchasing behavior was analyzed to uncover meaningful insights related to revenue generation, customer loyalty, subscription behavior, and product performance.

The project successfully transformed raw transactional data into actionable business insights and showcases practical skills required for Data Analyst roles, including data preparation, querying, visualization, reporting, and data storytelling.